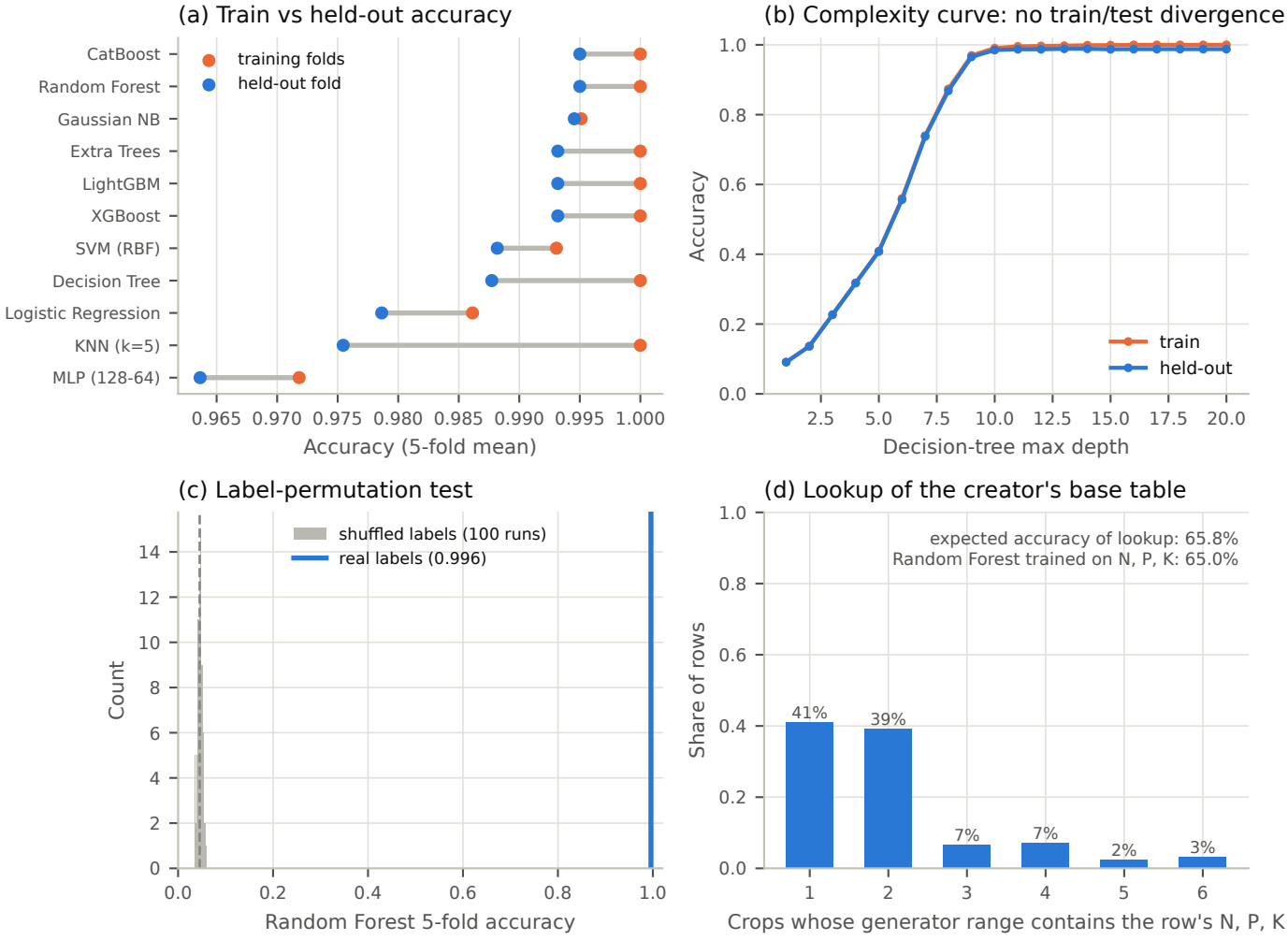
(a) Train vs held-out accuracy
CatBoost
Random Forest
Gaussian NB
Extra Trees
LightGBM
XGBoost
SVM (RBF)
Decision Tree
Logistic Regression
KNN (k=5)
MLP (128-64)
training folds
held-out fold
0.965 0.970 0.975 0.980 0.985 0.990 0.995 1.000
Accuracy (5-fold mean)
(b) Complexity curve: no train/test divergence
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
2.5 5.0 7.5 10.0 12.5 15.0 17.5 20.0
Decision-tree max depth
train
held-out
(c) Label-permutation test
shuffled labels (100 runs)
real labels (0.996)
Count
14
12
10
8
6
4
2
0
0.0 0.2 0.4 0.6 0.8 1.0
Random Forest 5-fold accuracy
(d) Lookup of the creator's base table
expected accuracy of lookup: 65.8%
Random Forest trained on N, P, K: 65.0%
Share of rows
1.0
0.8
0.6
0.4
0.2
0.0
41%
39%
7%
7%
2%
3%
1 2 3 4 5 6
Crops whose generator range contains the row's N, P, K